API Documentation

Student Cannabis Addiction Risk Prediction

COMP3011 · University of Leeds · 2025–2026

1. Overview

Base URL	`http://127.0.0.1:8000`
Protocol	`HTTP/1.1`
Request format	`application/x-www-form-urlencoded (form POST) or application/json`
Response format	`HTML (browser) — JSON extension planned for v2`
Authentication	`Django session auth (admin endpoints only)`
Framework	`Django 5.2 · Python 3.11`
Database	`SQLite3 (riskapp_predictionrecord table)`
ML Model	`Random Forest · scikit-learn · AUC-ROC 0.863 · Recall 0.916`
Dataset	`UCI Drug Consumption (Fehrman et al., 2015) — 1,885 participants`
Version	`rf_cannabis_v1`

2. Data Model — PredictionRecord

Every form submission creates one `PredictionRecord` row in the SQLite database. The table is the central data store for all CRUD operations exposed through the Django admin interface.

Field	Type	Constraints	Description
`id`	`INTEGER`	PK, auto-increment	Unique record identifier
`created_at`	`DATETIME`	auto_now_add=True	UTC timestamp of submission

answers_json	JSON	NOT NULL	12 UCI feature values as key-value pairs
risk_score	FLOAT	0.0 – 1.0	RF predicted probability for at-risk class
risk_band	VARCHAR(20)	choices enforced	low / moderate / high / very_high
model_version	VARCHAR(50)	default: rf_cannabis_v1	Model identifier for audit trail

risk_band values and probability boundaries (Random Forest threshold = 0.45):

Band	Probability	Recommended action
low	prob < 0.30	No immediate concern — provide awareness resources
moderate	0.30 ≤ prob < 0.55	Worth monitoring — suggest wellbeing check-in
high	0.55 ≤ prob < 0.75	Counselling referral advised
very_high	prob ≥ 0.75	Prioritise urgent intervention

3. Endpoints

3.1 Endpoint Summary

Method	Path	Operation	CRUD	Auth
GET	/risk/	Render assessment form	**Read**	None
POST	/risk/	Submit form, generate prediction	**Create**	None
GET	/admin/riskapp/predictionrecord/	List all predictions	**Read**	Admin
GET	/admin/riskapp/predictionrecord/<id>/	Retrieve single prediction	**Read**	Admin
POST	/admin/riskapp/predictionrecord/<id>/change/	Update a prediction	**Update**	Admin
POST	/admin/riskapp/predictionrecord/<id>/delete/	Delete a prediction	**Delete**	Admin

3.2 Endpoint Detail

GET `/risk/` *Render the assessment questionnaire*

Auth: None **Success:** HTTP 200 OK — HTML page

Returns the HTML questionnaire page containing three sections: Demographics (Age, Gender, Education, Country, Ethnicity), Personality Traits (Neuroticism, Extraversion, Openness, Agreeableness, Conscientiousness), and Behavioural Tendencies (Impulsivity, Sensation-Seeking). No authentication required.

Error Codes

Code	Meaning	When raised
404	Not Found	URL is not registered in urls.py
500	Server Error	ml_utils.py failed to load the model on startup

POST `/risk/` *Submit form and generate risk prediction*

Auth: None (CSRF token required) **Success:** HTTP 200 OK — HTML result page

Accepts the completed questionnaire form, maps all twelve feature values to UCI-quantified floats, passes them to the Random Forest classifier, saves the result to the database as a PredictionRecord, and renders the result page showing the risk band, probability score, animated gauge bar, and contextual support resources. CSRF token is required.

Parameters

Parameter	Type	Required	Description
`Age`	float	Yes	UCI-quantified age value (e.g. -0.95197 for 18–24)
`Gender`	float	Yes	UCI-quantified gender (Female: 0.48246, Male: -0.48246)
`Education`	float	Yes	UCI-quantified education level (-2.43591 to 1.98437)
`Country`	float	Yes	UCI-quantified country code
`Ethnicity`	float	Yes	UCI-quantified ethnicity code
`Nscore`	float	Yes	Neuroticism mapped to UCI range (-3.46436 to 3.46436)
`Escore`	float	Yes	Extraversion mapped to UCI range (-3.27393 to 3.27393)
`Oscore`	float	Yes	Openness mapped to UCI range (-3.27366 to 2.90161)
`Ascore`	float	Yes	Agreeableness mapped to UCI range (-3.46436 to 3.46436)
`Cscore`	float	Yes	Conscientiousness mapped to UCI range (-3.46436 to 3.46436)
`Impulsive`	float	Yes	Impulsivity mapped to UCI range (-2.55524 to 2.90161)
`SS`	float	Yes	Sensation-Seeking mapped to UCI range (-2.07848 to 1.92173)
`csrfmiddlewaretoken`	string	Yes	Django CSRF token (auto-injected by form)

Request Body

```
Content-Type: application/x-www-form-urlencoded csrfmiddlewaretoken=abc123&
Age=-0.95197&Gender=-0.48246&Education=1.98437& Country=0.96082&Ethnicity=0.11308&
Nscore=0.84580&Escore=-0.30172&Oscore=1.43533& Ascore=-0.80615&Cscore=-0.01928&
Impulsive=0.59042&SS=1.30612
```

Example Response

```
<div class="risk-hero" data-tier="very high"> <div class="risk-label">Very High Risk</div> <div
class="risk-prob"> Risk probability: <strong>0.905</strong> / 1.000 </div> </div>
```

Error Codes

Code	Meaning	When raised
403	Forbidden	CSRF token missing or invalid
500	Server Error	Model inference failed or database write failed

GET `/admin/riskapp/predictionrecord/` *List all prediction records (CRUD — Read)*

■ Auth: Required (Django admin superuser) **Success:** HTTP 200 OK — HTML admin list page

Django admin list view. Displays all PredictionRecord entries ordered by created_at descending. Supports filtering by risk_band and searching by model_version. Requires superuser login. Access the admin panel at /admin/ and log in with your superuser credentials.

Error Codes

Code	Meaning	When raised
302	Redirect	Unauthenticated request — redirected to /admin/login/
403	Forbidden	Authenticated but not a superuser

GET `/admin/riskapp/predictionrecord//` *Retrieve single prediction record (CRUD — Read)*

■ Auth: Required (Django admin superuser) **Success:** HTTP 200 OK — HTML admin detail page

Django admin detail view for a single PredictionRecord. Displays all fields including the full answers_json payload, risk_score, risk_band, created_at timestamp, and model_version.

Parameters

Parameter	Type	Required	Description
id	integer	Yes	Primary key of the PredictionRecord to retrieve

Error Codes

Code	Meaning	When raised
302	Redirect	Unauthenticated — redirected to /admin/login/
404	Not Found	No PredictionRecord with the given id exists

POST `/admin/riskapp/predictionrecord//change/`

Update a prediction record (CRUD — Update)

■ Auth: Required (Django admin superuser)

Success: HTTP 302 Redirect — back to list view on success

Django admin change view. Allows a superuser to edit any field of a PredictionRecord — for example, correcting a risk_band or updating the model_version tag. Submitting the change form sends a POST request to this endpoint.

Parameters

Parameter	Type	Required	Description
`id`	integer	Yes	Primary key of the record to update
`risk_band`	string	No	Updated risk band value
`model_version`	string	No	Updated model version tag
`answers_json`	JSON	No	Updated feature values (JSON string)

Error Codes

Code	Meaning	When raised
302	Redirect	Unauthenticated — redirected to /admin/login/
404	Not Found	No PredictionRecord with the given id exists
200	Form re-rendered	Validation error — form shown again with error messages

POST `/admin/riskapp/predictionrecord//delete/`

Delete a prediction record (CRUD — Delete)

■ **Auth: Required (Django admin superuser)**

Success: HTTP 302 Redirect — back to list view on success

Django admin delete view. Permanently removes a PredictionRecord from the database. Django admin shows a confirmation page on GET; the deletion is performed on POST confirmation. This operation is irreversible.

Parameters

Parameter	Type	Required	Description
`id`	integer	Yes	Primary key of the PredictionRecord to delete

Error Codes

Code	Meaning	When raised
`302`	Redirect	Unauthenticated — redirected to /admin/login/
`404`	Not Found	No PredictionRecord with the given id exists

4. Authentication

The API uses two distinct authentication mechanisms depending on the endpoint.

Endpoint	Auth method	How to authenticate
`GET /risk/`	None required	Publicly accessible. No credentials needed.
`POST /risk/`	CSRF token only	The Django CSRF middleware validates the csrfmiddlewaretoken field automatically injected by the form. No user login required.
`All /admin/ endpoints`	Django session (superuser)	Navigate to /admin/, enter superuser username and password. Session cookie is set automatically. Create a superuser with: python manage.py createsuperuser

5. Error Handling

Error handling in this API is distributed across four distinct layers. Each layer intercepts a different class of error before it can reach the next stage. Understanding which layer handles which error is important for debugging and testing.

5.1 Error Handling Layers

Layer	Handles	Response	Where defined
1 — Browser	Empty required fields (HTML5 validation)	Tooltip shown — POST never sent to Django	`required attribute in form.html`
2 — Django Middleware	Missing or invalid CSRF token	403 Forbidden — views.py never called	`CsrfViewMiddleware in settings.py`
3 — views.py	Missing POST field (KeyError) or non-numeric value (ValueError)	200 — form re-rendered with error message via Django messages	`try/except block in riskapp/views.py`
4 — Django Admin	Unauthenticated admin access / record not found	302 Redirect to login / 404 Not Found	`Built-in Django admin framework`
5 — ml_utils.py	Model file missing at startup	500 Internal Server Error — joblib.load() raises FileNotFoundError	`rf_model = joblib.load() in ml_utils.py`

5.2 Request Flow

```
Browser → CSRF Middleware → views.py → ml_utils.py → Database ↓ ↓ ↓ ↓ HTML5 403 if
```

```
200 + msg 500 if model tooltip no token if bad field file missing
```

5.3 HTTP Status Code Reference

Code	Status	Layer	When raised
200	OK	*Browser / views.py*	Successful page render, or form re-rendered after validation error
302	Found (Redirect)	*Django Admin*	Unauthenticated admin request redirected to /admin/login/, or successful admin save
403	Forbidden	*CSRF Middleware*	POST request missing csrfmiddlewaretoken, or token is invalid
404	Not Found	*Django Admin*	Admin URL not registered, or PredictionRecord with given id does not exist
500	Internal Server Error	*ml_utils.py*	rf_cannabis_model.joblib missing at startup, or unhandled exception in views.py

Note: HTML5 client-side validation (browser layer) prevents empty form submissions from reaching the server entirely — no HTTP status code is generated for this case as the POST request is never sent.

6. Input Feature Reference

The POST /risk/ endpoint accepts exactly 12 features. Demographic features are submitted as pre-mapped UCI float values selected from form dropdowns. Personality features are submitted as UCI-mapped floats computed from the 1–10 sliders by the form's JavaScript before submission.

Field	Source	Range	Description
Age	Dropdown	-0.95197 to 2.59171	18–24 through 65+
Gender	Dropdown	-0.48246 / 0.48246	Male / Female
Education	Dropdown	-2.43591 to 1.98437	Left school before 16 through Doctorate
Country	Dropdown	-0.57009 to 0.96082	Australia, Canada, NZ, Other, UK, Ireland, USA
Ethnicity	Dropdown	-1.65743 to 0.12600	Asian, Black, Mixed, Other, White
Nscore	Slider 1–10	-3.46436 to 3.46436	Neuroticism (Big Five)
Escore	Slider 1–10	-3.27393 to 3.27393	Extraversion (Big Five)
Oscore	Slider 1–10	-3.27366 to 2.90161	Openness to Experience (Big Five)
Ascore	Slider 1–10	-3.46436 to 3.46436	Agreeableness (Big Five)

Cscore	Slider 1–10	-3.46436 to 3.46436	Conscientiousness (Big Five)
Impulsive	Slider 1–10	-2.55524 to 2.90161	BIS-11 Impulsivity scale
SS	Slider 1–10	-2.07848 to 1.92173	Sensation-Seeking scale

7. Local Setup & Running the API

1. Clone the repository

```
git clone https://github.com/YOUR_USERNAME/addiction-risk-api.git cd
addiction-risk-api
```

2. Create and activate environment

```
conda create -n addiction-risk python=3.11 conda activate addiction-risk
```

3. Install dependencies

```
pip install django scikit-learn pandas joblib numpy
```

4. Generate the model

```
# Run all cells in: notebooks/random_forest_model.ipynb # This produces:
ml_artifacts/rf_cannabis_model.joblib
```

5. Apply migrations

```
python manage.py migrate
```

6. Create admin superuser

```
python manage.py createsuperuser
```

7. Start the server

```
python manage.py runserver
```

8. Access the application

```
Questionnaire: http://127.0.0.1:8000/risk/ Admin panel: http://127.0.0.1:8000/admin/
```

Dataset: UCI Drug Consumption (Quantified) — Fehrman, E. et al. (2015). arXiv:1506.06297. Available at: https://archive.ics.uci.edu/dataset/373/drug+consumption+quantified